

STUDENT NAME	MEMO
STUDENT NUMBER	

DEGREE	NDip Somatic Therapy
COURSE	General Pharmacology
COURSE CODE	GPA306D ✓
PAPER	Class Test 3
STANDARD / RE-EXAMINATION	Standard ✓
DATE	18 September 2026 ✓
TIME	09:00-10:30 ✓
MARKS	60 ✓
NUMBER OF STUDENTS	43 ✓
NUMBER OF PAGES (including this page)	7 ✓

INTERNAL EXAMINERS	MODERATORS
Mr KD Serabele	Prof E Olivier <i>[Signature]</i> 1 comment

SECTION	MARKS ALLOCATED	MARKS RECEIVED
A: Multiple Choice	15	
B: True and False	15	
C: Written Answers	30	
TOTAL	60	

INSTRUCTIONS TO CANDIDATES
1. Complete/attempt all the questions. Read all the questions carefully.
2. Answer all sections of the test in the spaces provided on the paper.
3. Negative marking will not be applied

SECTION A: Multiple Choice Questions

[15 Marks]

1. Circle each correct statement.
2. Each correct answer will receive plus ONE mark

Unit 15: Introduction to Microbiology.

(5 Marks)

1. What is the primary mode of reproduction for bacteria?

- a) Budding
- b) Spore formation
- c) Binary fission
- d) Sexual reproduction

2. Which of the following is a characteristic of prokaryotic cells?

- a) They have supercoiled DNA (genome).
- b) They have either DNA or RNA, not both.
- c) They have a rigid cell wall made of chitin.
- d) They are multicellular organisms.

3. What is the cell wall of a fungus primarily made of?

- a) Peptidoglycan
- b) Chitin
- c) Cellulose
- d) Keratin

4. Which of the following is an example of a yeast-like fungus?

- a) *Cryptococcus* spp.
- b) *Aspergillus*
- c) *Candida*
- d) *Trichophyton*

5. Which of the following is a pharmaceutically useful microbe used to validate and monitor moist heat sterilization processes?

- a) *B. subtilis*
- b) *Clostridium perfringens*
- c) *Actinomyces*
- d) *B. stearothermophilus*

1. Which line of defence is described as having "no memory" and being "non-specific"?

- a) First line of defence only.
- b) Both the first and second lines of defence.**
- c) Second line of defence only.
- d) Third line of defence only.

2. What is the function of B lymphocytes?

- a) They produce antibodies.**
- b) They kill by producing toxins.
- c) They regulate the activity of other cells.
- d) They scavenge dying body cells.

3. Which type of immune cells mediates the activity of cytotoxic cells?

- a) Macrophages
- b) B Cells
- c) Antibodies
- d) T Cells**

4. What is the cause of an autoimmune disease?

- a) The immune system overreacts to a harmless substance.
- b) A person is born with a weak immune system.
- c) The immune system attacks its own normal healthy cells.**
- d) A disease or infection weakens the immune system.

5. What is the process of a mother transferring antibodies to her child via the placenta or breast milk called?

- a) Natural passive immunity.**
- b) Natural active immunity.
- c) Artificial active immunity.
- d) Artificial passive immunity.

1. What is the primary difference between a disinfectant and an antiseptic??

- a) Disinfectants are used on living tissue, while antiseptics are used on non-living materials.
- b) Disinfectants are always more concentrated than antiseptics..
- c) Antiseptics are always more effective than disinfectants.
- d) Disinfectants are applied on non-living surfaces, while antiseptics are used to disinfect intact living tissue.

2. What is the key characteristic of sterilisation?

- a) It kills non-pathogens but not all pathogens.
- b) It does not eliminate bacterial spores.
- c) It is a complete removal or destruction of all living organisms.
- d) It is a process of removing vegetative microorganisms from inanimate objects.

3. What is a recommended preventative measure for a client who has a contra-indication for a treatment?

- a) Properly clean the affected area first.
- b) Give them a strong antiseptic.
- c) Only use disposable equipment.
- d) Stop the treatment immediately and refer them to a doctor.

4. What is the most effective method of heat sterilisation?

- a) Dry heat
- b) Moist heat (autoclaving)
- c) Ultraviolet radiation
- d) Boiling water

5. What is a key characteristic of low-level disinfection?

- a) It destroys all microorganisms.
- b) It has little or no sporicidal activity..
- c) It is used for instruments that come into contact with non-intact skin.
- d) It is achieved by means of sterilisation.

SECTION B: True/False Questions

[15 Marks]

1. Mark each statement as either **True** or **False** in the box provided to the left of the statement
2. Each correct answer will receive plus ONE mark

QUESTION 1:

Prokaryotic and Eukaryotic cells

Office
use only

A	Bacteria belonging to the Kingdom Monera.	T	
B	The cell wall of bacteria is made of chitin.	F	
C	Yeasts are a type of unicellular fungi.	T	
D	<i>It takes</i> Clostridium perfringens is a microbe used to confirm anaerobic growth conditions. <i>Overlap with C1.1</i>	T	
E	The purpose of staining cells on the microscope slide is to enlarge the cells.	F	
Total (1 X 5 = 5)			

QUESTION 2:

Control of Microbial Growth

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A	Sterilisation eliminates bacterial spores.	T	
B	A closed unventilated salon environment is a preventative measure against infection.	F	
C	Disinfectants have little or no sporicidal activity.	T	
D	The first step in controlling microbial growth is always sterilisation.	F	
E	High-level disinfection is used for instruments that only come into contact with intact skin.	F	
Total (1 X 5 = 5)			

QUESTION 3:

The Immune System

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use only

A	The first line of defense is always active and rapid.	T	
B	B cells are involved in antibody-mediated immunity	T	
C	The bone marrow is responsible for lymphocytes production.	T	
D	Hypersensitivity is when the immune system reacts against pathogens.	F	
E	Natural passive immunity is achieved through vaccination.	F	
Total (1 X 5 = 5)			

QUESTION 1: Pharmaceutical Uses of Microorganisms**[9]**

1.1 Indicate, while providing at least one example, how microorganisms play a critical role in both pharmaceutical manufacturing and quality assurance. (9)

Antibiotic production ✓

e.g. Actinomyces, Streptomycins, Penicillin ✓

Vaccine production ✓

e.g. MMR (measles, mumps and rubella); Corynebacterium diphtheria; Haemophilus influenza type b; Clostridium tetani; Rotavirus ✓

Validate- & monitor dry heat sterilisation method ✓

e.g. spore forming bacteria B. subtilis ✓

Validate and monitor moist heat sterilisation process ✓

e.g. B. stearothermophilus ✓

Confirm anaerobic growth conditions ✓ e.g. Clostridium perfringens ✓

QUESTION 2: Control of Microbes**[13]**

2.1 Differentiate between Sanitation, Sterilisation and Disinfection.

(6)

Sanitisation:

Remove debris ✓, germs – cleaning ✓

Sterilisation

A complete removal ✓ or destruction of all living organisms ✓

Disinfection:

Inhibition ✓ of microbes on non-living materials ✓

2.2 Describe the three methods of sterilisation.

(7)

Heat ✓ – using moist ✓ (autoclaving = most effective) or dry heat ✓

Chemical ✓ – using chemosterilants ✓

Radiation ✓ – using Ultraviolet/ Gamma ✓

QUESTION 3: Immune Disorders

[8]

3.1 What are the main differences between Innate and Adaptive immunity

Innate:

(4)

Always active✓✓

Rapid✓

Non- specific defenses✓

Has no memory✓

Adaptive:

(4)

Recognition✓

Attack✓

Memory✓

Specificity✓

SECTION A: 15 Marks

SECTION B: 15 Marks

SECTION C: 30 Marks

GRAND TOTAL: 60 Marks